Statement of Purpose (SOP) – Amazon ML Summer School 2026

My involvement with machine learning started with the passion of working with systems which can learn from data and solve real-world problems. This led to me getting interested in learning about Artificial Intelligence and Machine Learning outside of my college studies. As a Computer Science Engineering third-year undergraduate student, I have been actively exploring AI and ML through my courses, self-studies, and projects solving real-world problems.

One of the most interesting projects I have done is called ToxicGuard. It is an NLP-based project which helps detect toxic comments or cyberbullying content. This project was created to deal with the problem of cyberbullying or abuse content present on the social media website. I helped out with the data preprocessing, text cleaning, feature engineering, training, testing, evaluation, and deploying models for this project. Multiple data sets of toxic content available were joined together and vectorized using TF-IDF vectorization techniques. Various machine learning algorithms such as Logistic regression, Support Vector Machines, Naive Bayes classifier, and Random Forest were tested to find the best algorithm. After training all the machine learning models, the results obtained were around 99% accuracy in terms of predicting harmful or toxic content.

Apart from NLP, I have created a Heart Disease Prediction System using classification algorithms for predicting the probability of heart diseases based on the given dataset. It has increased my understanding of EDA, feature selection, comparison of different models, and application of machine learning in the field of healthcare. Moreover, I have enrolled in the Internshala Machine Learning training program to improve my knowledge about supervised learning, model evaluation, and data pre-processing in machine learning tasks.

From the above-mentioned experiences, I have gained enough practical exposure in the field of machine learning, but I feel the necessity of acquiring knowledge related to probability, statistics, linear algebra, optimization techniques, and deep learning. I am interested to learn about how advanced machine learning algorithms/models work and are implemented to address real-world problems on large scales.

The Amazon ML Summer School would be a great opportunity to learn from experienced machine learning scientists and researchers who have built and implemented machine learning solutions used by millions of people around the globe.

I think that having hands-on experience on NLP and Predictive Analytics projects, coupled with eagerness to learn and use new AI technologies, puts me in an advantageous position when it comes to applying to this program. In the long run, I would like to become an AI-driven Product Engineer capable of creating intelligent solutions which may change people’s life in a positive way. I am absolutely convinced that participation in Amazon ML Summer School will greatly contribute to achieving this goal.